Rules for differentiation

1

a $\frac{dy}{dx} = 6x$

b $\frac{dy}{dx} = 24x^3$

c $\frac{dy}{dx} = -12x^2$

d $\frac{dy}{dx} = 25x^4$

e $\frac{dy}{dx} = -\frac{3}{5}x^2$

f $\frac{dy}{dx} = 9x^5$

g $\frac{dy}{dx} = \frac{5x^3}{3}$

2

a $\frac{dy}{dx} = -\frac{21}{x^{\frac{5}{2}}}$

b $\frac{dy}{dx} = -\frac{3}{4x^4}$

3

a $\frac{dy}{dx} = 1$

b $\frac{dy}{dx} = 2$

c $\frac{dy}{dx} = \frac{2}{9}$

d $\frac{dy}{dx} = 2x - 1$

e $\frac{dy}{dx} = 2x - 8$

f $\frac{dy}{dx} = 4x^3 + 5x^4$

g $\frac{dy}{dx} = 5x^4 - 4x^3$

h $\frac{dy}{dx} = \frac{5x^4}{2} + \frac{8x^7}{5}$

i $\frac{dy}{dx} = x^7 + x^4 - 3$

4

a

i $y = -\frac{16x}{9} - \frac{32}{9}$

ii $\frac{dy}{dx} = -\frac{16}{9}$

b

i $y = 6x^2 + 23x + 15$

ii $\frac{dy}{dx} = 12x + 23$

c

i $y = 14x^3 + 4x^2$

ii $\frac{dy}{dx} = 42x^2 + 8x$

d

i $y = 15x^3 + 38x^2 + 24x$

ii $\frac{dy}{dx} = 45x^2 + 76x + 24$

e

i $y = x^2 + 8x + 16$

ii $\frac{dy}{dx} = 2x + 8$

f

i $y = 5x^2 - 30x + 45$

ii $\frac{dy}{dx} = 10x - 30$

g

i $y = 64x^2 - 64x + 16$

ii $\frac{dy}{dx} = 128x - 64$

5

a $\dfrac{dy}{dx} = 42x + 32$

b $\dfrac{dy}{dx} = 2x + 1$

Gradients

6 $f'(2) = 22$

7 $f'(4) = 0$

8 $f'(-4) = 2$

9 $f'(0) = \dfrac{1}{2}$

10

a 1

b $x = 1$

11 $f'(2) = 39$

12

a $f'(x) = 12x + 5$

b $f'(2) = 29$

c $x = 3$

13

a $f'(x) = 3x^2 - 4$

b $f'(4) = 44$

c $f'(-4) = 44$

d $x = \pm 5$

14

a $\dfrac{dy}{dx} = 4x - 8$

b $x = 2$

15 $x = \pm 3$